

Introduction to Quant Finance: Assignment-2

Folder Q1/

Holds the file `Q1-text.txt` which holds the stock selection info for this assignment.

Folder Q2/

Holds the folders `Data-Main/` and `Data-Split/` along with the file `Q2-Code.py`.

- Folder `Data-Main` holds the 4 files for the stock price data of our 4 selected stocks.
 - Folder `Data-Split` holds further two folders `Test/` - `Train/` where each of the two folder holds four CSV holding the stock price data of the stocks but splitted for 10-5 year for train-test respectively.
 - File `Q2-Code.py` holds the Code to download the stock price date via yfinance.
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Folder Q3/

Holds the 5 folders `1-HDFC/` , `2-INFOSYS/` , `3-MARUTI/` , `4-SUNPHARMA/` , `5-Patterns/` along with a file `Q3-Explained.pdf`.

- 4 folders `1-HDFC/` , `2-INFOSYS/` , `3-MARUTI/` , `4-SUNPHARMA/` are identical in struct. Each holds multiple chart images corresponding to Resistance(`Res-x.png`), Support(`Sup-x.png`), Sideways(`Side-x.png`) and Uptrend(`Up-x.png`).
 - 5th Folder `5-Patterns/` holds the 6 images corresponding to DoubleTop-Double-Bottom(`DBot-1.png`), DoubleTop(`DTop-1.png` , `DTop-2.png`), Head-Shoulder(`Hs-1.png` , `Hs-2.png`), Ranging Ascending(`RangeAsc.png`) patterns as asked in the last part of this assignment
 - Final file `Q3-Explained.pdf` explains the above folders in a fine way.
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Folder Q4/

Holds the Folder `Q4-Trades/` along with the files `Q4-Code.py` , `Q4-Metrics.pdf` , `Q4-Report.pdf` , `Q4-Trades.pdf` .

- File `Q4-Code.py` holds the simulation code for the question 4 where we try to make automated trades on the past data along with basic backtesting.
 - File `Q4-Metrics.pdf` have the comparision of 4 stocks with the evaluation metrics for test vs train.
 - File `Q4-Report.pdf` tried to explain the strategy and result evalutation of the Q4 in a simple way.
 - File `Q4-Trades.pdf` holds the table lisitng all the trades which were automatically taken duting the train-test on the past data of our 4 stocks.
 - Folder `Q4-Trades/` holds the 8 CSV files of 2 format `trainSMT` - `testSMT` of 4 each which represent the file holding the records of the trades taken by our snippet while train and test respectively. Files Itself are self explanatory there.
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File Assignment2.pdf

Holds the exact question proposed by the prof for the assignment.

File README.md

This file holding the info about the data in this whole DIR.

File README.pdf

PDF version of this file to simplify the stuffs.

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